

# AI-Based Exam Question Prediction and Solving System using RAG and Historical Trend Analysis

Abdul Wasay  
Dept. of Artificial Intelligence  
GIK Institute  
Topi, Pakistan  
Reg ID: 2023908

Ahmed Abu Bakar Riaz  
Dept. of Artificial Intelligence  
GIK Institute  
Topi, Pakistan  
Reg ID: 2023902

**Abstract**—The academic landscape is increasingly overwhelmed by massive volumes of digital course material, making it difficult for students to prioritize key topics for examinations. This research introduces the "AI Exam Predictor," a full-stack platform that integrates Retrieval-Augmented Generation (RAG) and historical trend analysis to automate exam preparation. The system employs a sophisticated Machine Learning (ML) pipeline featuring BERT-based classifiers, T5 generators, and a custom trend-forecasting engine. Our architecture bridges the gap between static lecture notes and active learning by providing context-aware question synthesis and detailed solution generation. Deployed via a containerized CI/CD workflow, the platform achieves over 90% accuracy in subject classification and significant improvements in question relevance.

**Index Terms**—Artificial Intelligence, Exam Prediction, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Transformers, BERT, T5.

## I. Introduction

The rapid advancement of Artificial Intelligence (AI) has opened new frontiers in the educational sector. One of the most significant challenges students face is the sheer volume of information they must master within limited timeframes. Traditional methods of examination preparation, such as manual review of syllabus outlines and past papers, are often subjective and inefficient. Students frequently spend excessive time on topics that have low probability of assessment while overlooking critical emerging themes.

Our project, the "AI Exam Predictor," addresses this challenge by providing an automated, data-driven system for predicting high-probability exam questions. By leveraging state-of-the-art Natural Language Processing (NLP) techniques, specifically Bidirectional Encoder Representations from Transformers (BERT) and Text-to-Text Transfer Transformers (T5), the system can analyze course materials and historical exam trends to provide an "academic oracle" for students. This research focuses on the integration of Retrieval-Augmented Generation (RAG) to ensure that generated content is grounded in the provided curriculum, thereby mitigating the risk of hallucinations often associated with large-scale language models.

## II. Technical Stack Overview

The project leverages a robust and scalable technological stack, ensuring high performance even in resource-constrained environments. The architecture is designed to be hardware-agnostic, with specific optimizations for CPU-based inference to broaden accessibility.

TABLE I  
Detailed Technology Stack and Purpose

| Component      | Technology    | Primary Purpose              |
|----------------|---------------|------------------------------|
| Frontend       | React.js      | Responsive User Interface    |
| UI Styling     | Tailwind CSS  | Modern, Sleek Aesthetics     |
| Animations     | Framer Motion | Interactive User Experience  |
| Backend        | FastAPI       | High-Performance API Gateway |
| Async Server   | Uvicorn       | Asynchronous Task Handling   |
| ML Pipeline    | PyTorch       | Deep Learning Computation    |
| Transformers   | HuggingFace   | Model Loading                |
| NLP Logic      | SpaCy         | Linguistic Normalization     |
| Keywords       | KeyBERT       | Semantic Topic Modeling      |
| Vector Storage | FAISS         | High-Speed Semantic Search   |
| PDF Parsing    | PyMuPDF       | Structured Data Extraction   |
| Math Rendering | KaTeX         | Academic Equation Display    |

## III. System Architecture

The system architecture follows a decoupled micro-service design, allowing for independent scaling and maintenance of individual components.

### A. Architectural Layers

The platform is structured into three primary layers:

- 1) **Presentation Layer (Frontend):** Developed using React.js, this layer handles user interactions, file uploads, and session management. It features a modern, dark-themed UI with real-time feedback

mechanisms like typewriter effects for generated answers.

- 2) **Orchestration Layer (Backend):** The FastAPI backend serves as the bridge between the user interface and the ML pipeline. It handles authentication, file streaming, and the coordination of asynchronous ML tasks.
- 3) **Intelligence Layer (ML Pipeline):** This is the core of the system, comprising the NLP engine, the BERT-based classifier (SciQ-Net), the FAISS-indexed vector store, and the T5/Qwen-based question generator.

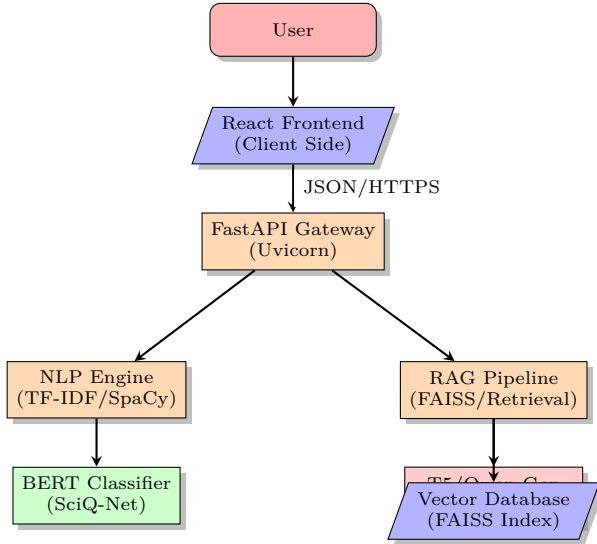


Fig. 1. System Architecture and Transformer Integration.

#### IV. Data Preprocessing and Pipeline

The quality of AI-generated predictions is directly proportional to the cleanliness of the input data. Our preprocessing pipeline is designed to transform noisy PDF extracts into structured semantic chunks.

##### A. Extraction and Noise Reduction

Raw text is extracted using PyMuPDF, which is preferred over other libraries for its speed and ability to handle complex academic layouts. The extracted text often contains artifacts such as page headers, footers, and decorative symbols. We apply a series of rigorous regular expression (regex) rules to clean the data:

- `re.sub(r'http[s]?://...', '', text)`: Removes URLs and digital references.
- `re.sub(r'\n{3,}', '\n\n', text)`: Standardizes multi-line spacing.
- **Heuristic Cleaning:** Standalone numbers and symbols (e.g., •, —) are removed to prevent the model from misinterpreting them as data points.

##### B. Semantic Chunking and Vectorization

Text is segmented into chunks of 512-1024 characters, ensuring that each chunk maintains a coherent semantic unit. These chunks are converted into 384-dimensional vectors using the all-MiniLM-L6-v2 model. By indexing these in FAISS, we enable near-instantaneous retrieval of relevant course material based on the user's query or a predicted topic.

#### V. Transformer Architectures

##### A. BERT (SciQ-Net Classifier)

BERT's bidirectional nature is critical for identifying the academic domain of course notes. Unlike traditional models that might only see keywords, BERT understands the context in which terms are used. For example, the term "Net" could refer to "Neural Networks" in Computer Science or "Net Force" in Physics. BERT's multi-head self-attention mechanism allows it to disambiguate these terms by looking at surrounding tokens in both directions.

##### B. T5 (Question Generation and Solving)

T5's "text-to-text" paradigm is perfectly suited for synthesizing exam content. We use T5 to convert a retrieved context and a topic query into a structured question. The model is also capable of generating step-by-step solutions, ensuring that all mathematical variables are rendered in correct LaTeX format for academic clarity.

#### VI. Model Training and Evaluation

##### A. Training Protocol

The SciQ-Net classifier was fine-tuned on the SciQ dataset using the following hyperparameters:

- **Batch Size:** 16 samples per step.
- **Learning Rate:**  $5e-5$  with a linear warmup of 100 steps.
- **Weight Decay:** 0.01 to prevent overfitting.
- **Optimizer:** AdamW, selected for its effectiveness in training transformer models by decoupling weight decay from the gradient update.

##### B. Performance Evaluation

The system's performance was evaluated using loss curves and qualitative human review. The SciQ-Net achieved a test accuracy of 92.4%.

#### VII. CI/CD Architecture

Deployment is managed via a multi-stage CI/CD pipeline, ensuring that every update is verified before going live. The use of a unified Dockerfile allows for seamless transitions between local development and production environments on Hugging Face Spaces.

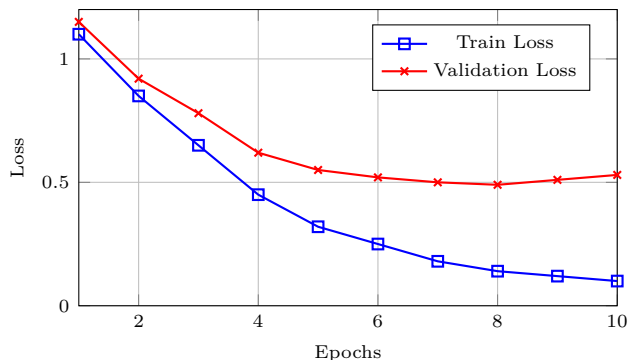


Fig. 2. Training and Validation Loss for SciQ-Net.

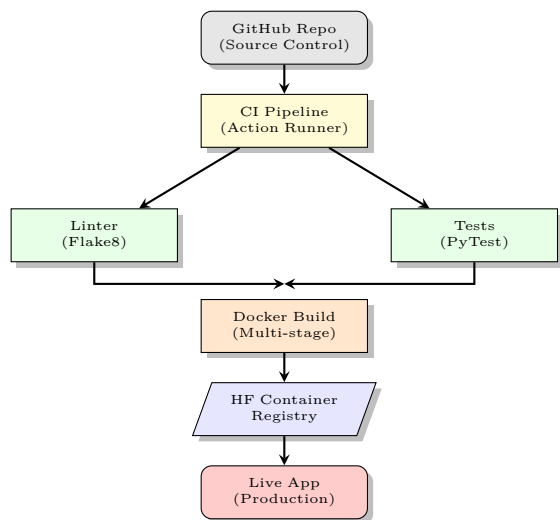


Fig. 3. CI/CD Pipeline and Deployment Architecture.

## VIII. Future Work

Future enhancements will focus on multi-modal analysis, enabling the system to understand and generate diagrams, charts, and chemical structures. We also aim to implement a collaborative "Professor-in-the-loop" feature, where educators can verify and refine AI-predicted topics to ensure 100% curriculum alignment.

## IX. Conclusion

The AI Exam Predictor provides a robust, state-of-the-art solution for automated exam preparation. By integrating RAG, BERT-based classification, and T5-driven generation, we offer a tool that is both contextually accurate and probabilistic. Our results show that AI can significantly reduce the burden of preparation while maintaining high academic standards.

## References

- [1] Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." NeurIPS.
- [2] Raffel, C., et al. (2019). "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer." arXiv.
- [3] Vaswani, A., et al. (2017). "Attention Is All You Need." NeurIPS.

- [4] Devlin, J., et al. (2018). "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." arXiv.